

Open Data Exploration & Visualization Assignment

In this assignment, you will design and build an interactive data application that showcases your data analysis skills. You are free to choose your own dataset and decide how to explore, clean, analyze, model, and visualize the data. The goal is not to build the most complex system, but to demonstrate clear thinking, technical competence, and good data practices.

Objective

Build an interactive data application (using Streamlit) that allows a user to explore a dataset, gain insights, and understand patterns or relationships in the data. Your application should clearly demonstrate your skills in Python, pandas, data visualization, and basic data science workflows.

Dataset

You may choose any dataset you like, provided that:

- The dataset is non-trivial (at least a few hundred rows or multiple variables)
- The data can reasonably be explored, cleaned, and visualized
- Public datasets (Kaggle, data.gov, Open Data portals, etc.) are allowed
- You may also use your own collected data

You must clearly state the source of your dataset in your application or documentation.

Core Requirements

Your application should include the following components:

1. Data loading

- Load data from a file (e.g. CSV)
- Handle basic data types correctly

2. Data cleaning

- Handle missing values in a reasonable way
- Demonstrate at least one automatic or rule-based cleaning step

3. Exploration & statistics

- Show descriptive statistics
- Allow filtering or slicing of the data

4. Visualizations

- Create multiple meaningful visualizations
- Use Matplotlib and/or Seaborn
- Visualizations should support understanding, not decoration

Optional Extensions (Choose Freely)

You are encouraged to extend your application in any direction you find interesting.

Examples include:

- Add geographical visualizations (maps, coordinates, regions)
- Detect outliers or anomalies
- Create a small data story or insight summary
- Allow users to download filtered or processed data
- Build a small machine learning model (regression or classification)
- Show model performance using appropriate metrics

These extensions are optional but can help demonstrate deeper understanding.

Freedom & Creativity

This is a deliberately open assignment. There is no single correct solution.

You are expected to:

- Make your own design choices
- Explain why you made those choices
- Focus on clarity, structure, and usability

A simple but well-thought-out project is better than a complex but unclear one.